A computational model of the evolution of honeybee waggle-dance communication

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1 Introduction

Honeybee waggle dances provide spatial information about resources away from the nest. One ecological hypothesis is that such communication is favored when food is difficult to discover independently but remains useful after a worker has found it. Empirical and theoretical studies suggest that the value of dance information and recruitment depends on the spatiotemporal distribution of food resources, including patch density, patch size, reward variability, and habitat [7, 3, 4, 1, 2, 6, 5]. The model is deliberately minimal: it asks which evolutionary pressures are needed before adding richer social, developmental, or landscape detail. We start from direct directional communication on an exposed horizontal comb, where dance angle can point in the food direction, and then ask what additional conditions make a gravity-referenced vertical-comb code reachable.

The simulations evaluate four evolutionary hypotheses.

1. **Resource-distribution hypothesis.** Costly directional communication should be favored when independent discovery is difficult enough that recruitment is useful, but not so difficult that few dances are seeded. It should be weakly favored when food is very easy to find without dances.

2. **Architectural-precondition hypothesis.** Direct pointing should work well on a horizontal comb, whereas gravity-referenced communication should become useful only when the comb geometry supplies a reliable gravity cue. A non-communication benefit of vertical combs may therefore be needed to make the gravity channel selectively available.

3. **Sender–receiver coordination hypothesis.** Sender and receiver transposition must evolve together. Positive coupling between sender and receiver mutations should reduce the time spent in mismatched codes.

4. **Transition-risk hypothesis.** Selection for vertical combs can outpace adaptation of the communication code. Successful transitions should therefore require overlap among vertical geometry, directional precision, receiver attention, and matched sender–receiver transposition; otherwise foraging may collapse before the new code becomes useful.

2 Model

2.1 Conceptual model

Colonies are the reproducing entities, while workers are behavioral agents sampled from colony-level trait means. A colony has seven heritable mean traits: directional precision investment, receiver attention, sender transposition, receiver transposition, search limit, comb tilt, and comb orientation. Directional precision investment is a costly investment in producing directionally informative dances. Receiver attention controls whether workers use available dances. Sender and receiver transposition interpolate between two communication modes: direct pointing in the comb plane and sun/gravity-referenced encoding. Comb tilt and orientation define the dance plane in the world, and search limit determines how far a worker searches along its chosen direction.

Each foraging episode samples food sites with direction, distance, angular width, value, and capacity. Workers act sequentially. If dances are available, a worker may follow one; otherwise it searches independently in a random direction. Independent searchers that find food add a noisy dance for that site. There are therefore no fixed scouts in the model. Background discovery and recruitment both emerge from the food distribution, receiver attention, search limits, and dance precision. This follows ecological recruitment models in which communication is valuable only when resources are discoverable and worth recruiting to [4, 1].

Comb geometry determines which directional cue is available. Direct pointing projects the horizontal food direction onto the comb plane, so its strength depends on the food direction, comb tilt, and comb orientation. Gravity- referenced mapping uses the projection of gravity onto the comb plane together with the episode's sun azimuth. Gravity provides no in-plane reference on a horizontal comb and becomes a stronger cue as the comb becomes vertical. Comb orientation can be treated as circular, or as axial when orientations ϕ and $\phi + 180^\circ$ represent the same comb plane.

2.2 Mathematical implementation

Let a colony's mean strategy be

$$\theta = (b, a, s, r, \ell, t, \phi),$$

where $b \in [0, 1]$ is directional precision investment, $a \in [0, 1]$ is receiver attention, $s, r \in [0, 1]$ are sender and receiver transposition, $\ell \geq 0$ is search limit, $t \in [0, 1]$ is comb tilt, and ϕ is comb orientation. A worker i receives stable within-colony deviations from the colony means:

$$\theta_i = \text{clip}(\theta + \epsilon_i).$$

These deviations represent worker heterogeneity but are not themselves heritable.

For a food direction α and sun azimuth ψ, let $D(\alpha; t, \phi)$ be the direct projected dance angle and $G(\alpha, \psi; t, \phi)$ be the sun/gravity-referenced dance angle. Let q_D and q_G be their corresponding cue strengths in the comb plane. A sender with transposition s_i encodes the dance mean angle as a weighted circular mean,

$$\mu_i = \text{cmean}\left((D, (1 - s_i)q_D),\ (G, s_i q_G)\right).$$

The produced dance signal is

$$x_i = \text{VM}(\mu_i, \kappa_{\max} b_i) + \eta_d \pmod{2\pi},$$

where VM is the von Mises distribution, $\kappa_{\max} = 14$ in the reported simulations, and η_d is transient dance-production noise. A receiver with transposition r_j applies the analogous weighted decoding rule and adds interpretation noise η_r to obtain a search direction $\hat{\alpha}_j$.

A worker succeeds at food site k only if

$$\Delta(\hat{\alpha}_j, \alpha_k) \leq w_k \quad \text{and} \quad d_k \leq \ell_j,$$

where Δ is circular angular distance, w_k is site angular width, and d_k is site distance. If multiple available sites satisfy this condition, the nearest site is used.

The colony payoff for an episode is

$$P_e = \sum_{j\in F_e} v_{k(j)} - c_T \left(\sum_{j\in F_e} d_{k(j)} + \sum_{j\in U_e} \ell_j \right) - \sum_{i\in D_e} (c_0 + c_b b_i) - c_a A_e + \beta_V t.$$

Here F_e is the set of successful workers, U_e the unsuccessful workers, D_e the independent finders that produce dances, and A_e the number of dance-following attempts. The terms are food value, travel/search cost, per-dance communication cost, receiver-attention cost, and an optional non-communication benefit proportional to comb verticality. Mean colony payoff is averaged over episodes and floored at 0.001 so that low-payoff colonies remain selectable but rare.

Daughter colonies are sampled in proportion to payoff. Only colony means are inherited. Mutations are Gaussian and clipped to the allowed ranges; sender and receiver transposition mutations can have correlation ρ, which is the coupling parameter used in the vertical-transition probes.

2.3 Simulation parameters and reproducibility

Unless otherwise stated, reported runs use 60 colonies, 80 workers per colony, 50 foraging episodes per colony per generation, 12 foraging attempts per episode, ten random seeds, within-colony standard deviation 0.08 for unit-interval traits, mutation standard deviation 0.07, dance-production noise standard deviation 0.18, interpretation-noise standard deviation 0.12, maximum search distance 8.0, food value 1.0, food-site capacity 6, travel cost 0.02 per distance unit, precision-dependent dance-cost coefficient $c_b = 0.02$, baseline dance cost $c_0 = 0$, and attention cost $c_a = 0.01$. Search-limit noise and mutation use the same scales multiplied by the maximum search distance. Initial directional precision is sampled from $[0, 0.15]$, initial receiver attention from $[0, 0.25]$, initial transposition traits are 0, and initial search limit is sampled from 15–45% of the maximum search distance.

The tables and figures in this report are generated from tracked result files rather than hand-entered into the document. Running

```
python -u experiments/run_report_artifacts.py artifacts
```

rebuilds the LaTeX fragments from the CSV files in `results/`; running the same script with `all` first regenerates the result CSVs. The provenance manifest is `report/artifact_sources.csv`.

3 Results

3.1 Horizontal comb

We first compared three simple food distributions over ten random seeds (101–110). The "hard" condition had one narrow food site per episode, the baseline condition had two moderately wide food sites, and the "easy" condition had many broad food sites. Table 1 reports the generation at which mean directional precision investment first reached 0.30, plus final-generation trait, success, and payoff values.

Each run used 60 colonies, 80 workers per colony, 30 generations, 50 foraging episodes per colony per generation, 12 foraging attempts per episode, and horizontal-comb tilt $t = 0$. Food-site distances were sampled uniformly from 1.0 to 8.0, maximum search distance was 8.0, travel cost was 0.02 per distance unit, baseline dance cost was 0, the precision-dependent dance-cost coefficient was 0.02, and food-site value was 1.0. All three conditions used food-site capacity 6. The conditions varied only the number and angular width of food sites: hard used one site of width 0.08, baseline used two sites of width 0.20, and easy used eight sites of width 0.50. The hard condition therefore makes independent discovery unlikely, while the easy condition makes independent discovery likely even without using the dance.

Condition	Reached	Reach gen.	Prec.	Attention	Search	Success	Payoff
Hard	4/10	22.750	0.264	0.313	2.789	0.008	0.001
Baseline	10/10	7.700	0.833	0.869	7.215	0.192	0.690
Easy	5/10	19.600	0.279	0.369	7.067	0.708	7.366

Table 1: Food-distribution experiment under the current dynamic recruitment model. "Reached" is the number of seeds in which mean directional precision investment reached 0.30 by generation 30. "Reach gen." is averaged over reached seeds only. Prec. is final mean directional precision investment; attention, search limit, success, and payoff values are also final-generation means over all ten seeds.

The result supports the resource-distribution hypothesis. In the hard environment, food is so rare and narrow that independent discoveries seldom seed dances; final success and payoff remain close to zero, and communication crosses the directional-precision threshold in only 4 of 10 seeds. In the baseline environment, independent discovery is sufficient to seed recruitment cascades, and directional precision, receiver attention, and search limit all rise strongly. In the easy environment, success and payoff are high, but directional precision and receiver attention remain much lower than in the baseline case because food is often found without precise communication.

3.2 Vertical comb

We also compared colonies initialized with horizontal, tilted, and vertical combs while holding the baseline food distribution fixed. Initial comb tilt was set to $t = 0$, $t = 0.5$, or $t = 1.0$, with ten seeds per condition. Comb tilt was still heritable and mutable in these runs. Table 2 summarizes the result after 30 generations.

Comb	Reached	Reach gen.	Prec.	Attention	Sender	Receiver	Success	Payoff
Horizontal	10/10	7.700	0.833	0.869	0.190	0.181	0.192	0.690
Tilted	10/10	11.000	0.799	0.891	0.148	0.170	0.193	0.685
Vertical	10/10	13.700	0.698	0.824	0.486	0.489	0.177	0.522

Table 2: Comb-tilt comparison using the baseline food distribution and ten seeds. Sender and receiver denote final mean sender and receiver transposition traits.

With mutable tilt, all three initial-tilt conditions reach the directional-precision threshold in all ten seeds under the current baseline settings. Tilted and vertical starts retain substantial foraging success over 30 generations, but they also evolve higher sender and receiver transposition than horizontal starts. The initially vertical condition still has lower payoff than the horizontal condition, and its final mean tilt is reduced in the underlying result file, indicating that the short-run comparison does not by itself establish a stable vertical-comb transition.

3.3 Tilt-geometry calibration

We next ran a calibration grid over initial comb tilt and the vertical-comb benefit β_V. These runs used the baseline food distribution, the daytime sun arc from 0° to 180° centered at 90°, and ten seeds (101–110). Success is the fraction of foraging attempts that found food. Payoff is the final mean colony payoff after food rewards, search/travel costs, dance costs, attention costs, and the vertical-comb benefit. Table 3 shows that increasing β_V can preserve more vertical combs, but these settings produce a sharp tradeoff: strong verticality often comes with reduced foraging success.

Initial tilt	Benefit	Final tilt	Sender	Receiver	Success	Payoff	Reached
0.000	0.150	0.142	0.178	0.183	0.195	0.735	10/10
0.000	0.200	0.162	0.192	0.208	0.195	0.762	10/10
0.000	0.250	0.151	0.229	0.196	0.194	0.751	10/10
0.500	0.150	0.373	0.225	0.230	0.137	0.535	9/10
0.500	0.200	0.445	0.282	0.271	0.137	0.558	9/10
0.500	0.250	0.330	0.208	0.206	0.154	0.601	10/10
1.000	0.150	0.737	0.341	0.339	0.090	0.313	7/10
1.000	0.200	0.919	0.291	0.282	0.001	0.093	4/10
1.000	0.250	0.917	0.288	0.333	0.002	0.127	7/10

Table 3: Ten-seed calibration of the vertical-comb benefit under the explicit tilt-geometry model. Sender and receiver are final mean transposition traits. Reached is the number of seeds in which mean directional precision investment reached 0.30 by generation 30.

The calibration suggests that the transition is path-dependent. Starting from a horizontal comb, benefits in the range 0.15–0.25 do not move colonies far from horizontal within 30 generations. Starting from a tilted comb, the same benefit range can retain intermediate tilt but with lower success. Starting from a vertical comb, benefits around 0.20–0.25 retain high verticality, but foraging nearly fails in most seeds. Thus the model can impose substantial architecture pressure, but the 30-generation calibration may end before sender transposition, receiver transposition, and directional precision have enough time to adapt together. The relevant open question is therefore whether these cases represent stable collapse or a longer coevolutionary transition.

3.4 Tilt orientation

We also examined the evolved orientation of the comb tilt. The current sun arc is centered at 90°, so Table 4 reports the weighted mean orientation in degrees and its offset from this sun-arc center. The within-alignment column measures how strongly colonies within a run converge on an orientation. The across-alignment column measures whether different seeds converge on the same absolute orientation, weighting each seed by its within-run orientation alignment.

Orientation sometimes aligns within a run, but it does not show a clean, robust relationship to the daytime sun arc across seeds under the circular orientation representation. Some conditions point near the sun-arc center, some near the opposite direction, and several have weak across-seed alignment. This indicates weak or path-dependent selection on orientation in these probes and motivates the axial representation used in the longer transition runs, where ϕ and $\phi + 180°$ are treated as the same comb plane.

3.5 Constrained vertical coupling

The tilt-calibration results left open whether gravity-referenced communication is difficult because the sender–receiver code itself is hard to reach, or because colonies must evolve the vertical geometry and the sender–receiver code at the same time. To separate these possibilities, we ran a constrained

Initial tilt	Benefit	Final tilt	Success	Within align.	Mean orient.	Sun offset	Across align.
0.000	0.150	0.142	0.195	0.518	122.841	32.8	0.473
0.000	0.200	0.162	0.195	0.489	155.097	65.1	0.492
0.000	0.250	0.151	0.194	0.478	79.011	-11.0	0.084
0.500	0.150	0.373	0.137	0.490	158.945	68.9	0.249
0.500	0.200	0.445	0.137	0.356	214.421	124.4	0.158
0.500	0.250	0.330	0.154	0.523	73.900	-16.1	0.401
1.000	0.150	0.737	0.090	0.545	251.668	161.7	0.374
1.000	0.200	0.919	0.001	0.374	263.771	173.8	0.093
1.000	0.250	0.917	0.002	0.392	266.776	176.8	0.291

Table 4: Orientation sanity check under the explicit tilt-geometry model. Mean orientation and sun offset are measured in degrees. Across alignment is the weighted across-seed orientation alignment; values near 0 indicate that different seeds choose different absolute orientations.

probe with the geometric precondition supplied. Initial comb tilt was fixed near vertical at $t = 0.95$, comb-tilt mutation was set to 0, comb orientation was treated axially, the vertical-comb benefit was set to 0, and the baseline food distribution was otherwise retained. We then varied only the sender–receiver transposition mutation correlation and ran each condition for 120 generations over five seeds (101–105). A run was counted as reaching gravity-referenced communication when both mean sender and receiver transposition reached 0.50.

Corr.	Reached	Reach gen.	Prec.	Sender	Receiver	Gap	Success	Payoff
0.000	5/5	42.600	0.878	0.864	0.881	0.033	0.192	0.680
0.300	5/5	46.200	0.847	0.913	0.898	0.022	0.192	0.694
0.600	5/5	21.000	0.882	0.890	0.897	0.010	0.199	0.754
0.900	5/5	22.400	0.863	0.907	0.903	0.005	0.194	0.723
1.000	5/5	23.400	0.836	0.901	0.901	0.000	0.191	0.677

Table 5: Constrained near-vertical coupling probe. Corr. is the correlation between sender and receiver transposition mutation increments. “Reached” counts seeds in which both mean sender and receiver transposition reached 0.50 by generation 120. Reach generation is averaged over reached seeds only. Gap is the final mean absolute sender–receiver transposition difference.

Table 5 separates reachability from transition speed. With near-vertical geometry supplied, every correlation condition reaches gravity-referenced communication in all five seeds, so the gravity channel is not intrinsically unreachable. The main difference is how quickly the matched sender–receiver code appears. Independent or weakly coupled mutations ($\rho = 0$ or 0.3) reach the threshold only after about 43–46 generations on average. Moderate to strong coupling ($\rho = 0.6$–1.0) reaches it by about generation 21–23. The final sender–receiver gap also falls as coupling increases: from 0.033 without coupling to 0.010 at $\rho = 0.6$ and 0 at $\rho = 1.0$. Payoff peaks at $\rho = 0.6$, not at perfect coupling, suggesting that coupling helps align the code but complete lockstep is not necessarily the highest-payoff mutation structure in this finite-seed probe. The result supports the sender–receiver coordination hypothesis and implies that the harder problem is the joint transition in which comb architecture, orientation representation, and the sender–receiver code must all become useful together.

3.6 Long transition runs

We therefore ran longer transition experiments using the axial orientation representation. These runs used the baseline food distribution, 120 generations, ten seeds (101–110), sender–receiver transposition mutation correlation 0.6, and the same three initial comb tilts and vertical-comb benefits as the calibration grid. Unlike the constrained near-vertical probe, comb tilt was free to mutate. A seed was counted as reaching the gravity channel when both mean sender and receiver transposition reached 0.50. It was counted as retaining verticality when final mean comb tilt was at least 0.80. It was counted as collapsed when mean foraging success fell to 0.02 or below at any generation, and as recovered when a collapsed seed ended with final success at least 0.10.

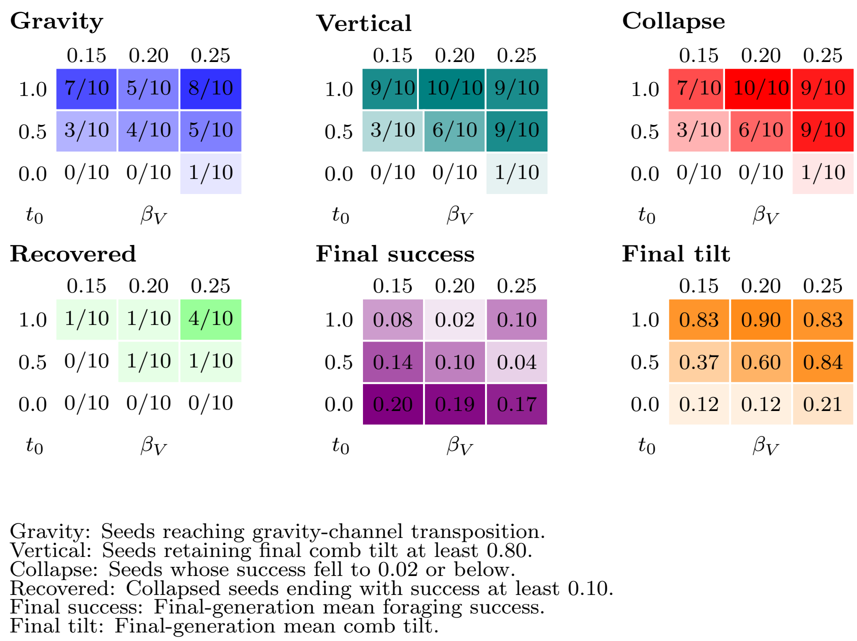

Figure 1: Long axial-orientation transition experiment. Rows are initial comb tilt, columns are vertical-comb benefit. Count panels report seeds out of ten; continuous panels report final-generation means. The generated fragment records the source CSV.

The longer horizon changes the interpretation but does not by itself solve the transition (Figure 1). From horizontal starts, weak and moderate vertical-comb benefits preserve high foraging success but do not move colonies far from horizontal. At $\beta_V = 0.25$, one seed reaches both verticality and the gravity channel, but that seed also collapses. From tilted starts, higher benefits increasingly retain verticality and allow more seeds to reach the gravity channel, but collapse also becomes common; at $\beta_V = 0.20$, one seed recovers to a high-success vertical gravity-channel state. From vertical starts, successful vertical gravity-channel outcomes are possible, especially at $\beta_V = 0.25$ where four collapsed seeds recover by generation 120, but most seeds still experience collapse. Thus the 30-generation failures were too short to distinguish an impossible transition from a slow coordination process, while 120 generations show that longer time alone is insufficient to make the transition robust under the current parameters.

4 Conclusion

The resource-distribution experiments support the idea that costly directional communication is favored most strongly in an intermediate ecological regime: food must be hard enough to make recruitment valuable, but discoverable enough for independent finders to seed dances. The geometry experiments support the architectural-precondition hypothesis in a more conditional way. A vertical-comb benefit can preserve verticality, and supplied near-vertical geometry allows gravity-referenced sender–receiver transposition to evolve, but the full architecture-and-code transition is not yet robust.

The constrained coupling probe supports the sender–receiver coordination hypothesis: with near-vertical geometry supplied, gravity-referenced communication evolves in every condition, and correlated sender–receiver mutations shorten the transition and reduce final mismatch. The long transition runs support the transition-risk hypothesis. Some tilted or vertical starts recover into high-success vertical gravity-channel states by generation 120, but horizontal starts mostly remain horizontal, and many vertical-retaining seeds lose foraging performance before communication catches up. The next modeling priority is therefore to calibrate vertical-comb benefits, mutation scales, and coupling assumptions so that architecture and communication can coevolve without routinely passing through a collapse in foraging success.

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